Final Project

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Course: CS573 – Data Visualization

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COVID 19 Global Interactive Dashboard

Problem Statement:

COVID-19 data is widely available, but existing dashboards often overwhelm users with raw numbers and complex visuals, making it difficult for non-experts to compare how the pandemic evolved across countries and continents or to see how vaccination trends relate to cases and deaths. This project addresses that problem by designing an interactive, population-aware dashboard that allows users to explore and compare COVID-19 cases, deaths, and vaccinations over time and across regions in a clear and interpretable way.

Introduction and Motivation

The COVID-19 pandemic generated a large amount of country-level data on cases, deaths, and vaccinations, but these numbers are often difficult for non-expert users to interpret when presented as isolated totals or overly crowded dashboards. Many pandemic dashboards emphasize snapshots or raw counts, which can make cross-country comparison misleading, especially when population size and timing of outbreaks differ substantially.

The main goal of this project is to design an interactive dashboard that helps users compare how the pandemic unfolded across countries and continents over time, while also exploring how vaccination patterns relate to cases and deaths. Rather than focusing on a single statistic, the dashboard combines a geographic view, a temporal comparison view, and country-level summary information so that users can move between global overview and detailed comparison in one interface.

The core problem addressed by this project is therefore a communication and comparison problem: how can COVID-19 data be presented in a way that is visually clear, population-aware, and useful for comparing countries with very different outbreak histories? To answer this, I created a dashboard that integrates a choropleth world map, a multi-country time-series chart, and a country detail panel with interactive controls for metric, continent, and country selection.

Related Work:

Several existing COVID-19 dashboards and design guidelines informed this project. The COVID-19 Data Explorer and other interactive pages from Our World in Data provided a key reference for how to present country-level cases, deaths, and vaccinations over time while allowing users to select multiple locations and switch between different metrics in one interface. These visualizations also emphasize transparent data sources, per-capita indicators, and the ability to download underlying datasets, which motivated the choice to use the same dataset and to focus on normalized measures such as cases and deaths per million and vaccinations per hundred in this project.

Research on public COVID-19 dashboards has identified concrete design principles that also shaped the dashboard developed here. A review of 100 U.S. city, county, and state COVID-19 dashboards derived ten best practices for public-facing health dashboards, including simple navigation, high usability, clear chart descriptions, and comparison views that allow users to relate one region to another. These recommendations directly support the decision to keep the interface to three coordinated views - a map, a multi-country time series, and a country detail panel - with straightforward controls for metric, continent, and country selection, rather than many separate pages or highly customized views. Practical guidance on COVID-19 data visualizations from Harvard's Data-Smart City Solutions further stresses explaining labels and terms clearly, showing trends over time instead of only daily snapshots, and being consistent about how metrics are counted, especially in a rapidly evolving pandemic. Those ideas reinforced the use of temporal line charts instead of single-day views, the inclusion of short explanatory text near the map and axes, and the consistent use of population-adjusted indicators across the dashboard.

Questions

This project grew out of a few simple but important questions about the pandemic and how to show its data clearly:

- **How did COVID-19 waves play out differently across countries and continents?**
  The first goal was to understand not just how many cases countries had, but when they experienced major waves and how intense those waves were. Some countries saw large early waves, others were hit later, and some had many smaller peaks instead of a few big ones. The dashboard answers this by letting users plot several countries on the same smoothed time-series chart, so it is easy to see whether two countries peaked at the same time, which one had higher peaks, and how long each wave lasted. By sharing the same horizontal (time) and vertical (value) scales, the chart turns raw dates and numbers into a clear visual story about how the pandemic evolved in different places.

- **Which countries were hit hardest once population size is considered?**
  Raw totals can be misleading because large countries will almost always have more cases and deaths simply due to having more people. A key question for this project was therefore: where was the impact proportionally highest? The dashboard addresses this using per-million indicators on the choropleth map, so each country is colored based on cases or deaths per million residents rather than raw counts. This makes it possible to

quickly pick out countries that have experienced a heavy burden relative to their size, including smaller nations that might otherwise be overlooked. The country detail panel reinforces this by showing both totals and per-capita numbers side by side, helping users understand the difference between "big numbers" and "big impact."

- **What happened to cases and deaths after vaccination campaigns started?**
  Another motivation was to help people visually connect vaccination data with the later course of the pandemic, without needing to run formal statistical models. The question here is whether users can see, for a given country, how the shape of case and death curves changes as vaccination coverage rises. The dashboard supports this by allowing users to switch the metric between cases, deaths, and vaccinations while keeping the map, time-series chart, and detail panel in sync. For example, a user can first view the vaccination curve for a country, then switch back to cases and compare the timing of peaks before and after large portions of the population were vaccinated. This does not prove causality, but it makes it much easier to explore patterns and ask informed follow-up questions.

- **How can all of this be shown in a dashboard that non-experts can understand and use?**
  Finally, the project asks a design question: how to present all this information without overwhelming people who are not data specialists. Many existing dashboards are powerful but can feel cluttered or confusing, especially when they expose many technical options at once. In response, this dashboard deliberately uses a small number of coordinated views maps for overview, a time-series chart for changes over time, and a detail panel for precise values - rather than many separate charts. Controls are kept simple: a metric dropdown, a continent filter, and a country selector that also responds to map clicks. Short explanatory text clarifies what colors and axes mean. Together, these choices aim to let a general user answer the earlier questions - about waves, relative burden, and vaccination patterns - by interacting directly with the visualization, instead of having to interpret long tables or advanced analytics.

Data Preprocessing:

This project uses the Our World in Data COVID-19 dataset, which provides regularly updated country-level information on confirmed cases, deaths, vaccinations, and related indicators over time. The dataset is suitable for this project because it includes both temporal and comparative variables, allowing analysis of pandemic trends across multiple countries and regions using a single consistent source.

For the dashboard, I selected variables that directly support the research goals, including date, location, continent, total cases, new cases smoothed, total deaths, new deaths smoothed, total cases per million, total deaths per million, people vaccinated, people fully vaccinated, and total vaccinations per hundred. I focused on these variables because they allow comparison of both absolute trends and population-adjusted impact, and because vaccination metrics can be explored alongside cases and deaths rather than in isolation.

The raw dataset was filtered to retain country-level records and remove aggregated entries that are less useful for country comparison. I then created smaller processed files for use in the dashboard: dashboard_data.csv for the time-series visualization, latest_snapshot.csv for country summaries and map coloring, continent_summary.csv for regional filtering support, and countries. geojson for the world map geometry. This preprocessing step improved performance and made the web dashboard easier to manage because the visualization loads only the fields required for interaction.

I also chose to emphasize per-million and per-hundred indicators where appropriate because raw totals alone can exaggerate the burden in highly populated countries. Using normalized measures makes the comparison more meaningful and helps users evaluate relative impact rather than simply comparing country size.

Exploratory Data Analysis (EDA):

Before building the final dashboard, I conducted exploration analysis to understand the structure of the data and identify which variables would be most informative for interactive comparison. Initial plots of cases and deaths over time showed that countries experienced pandemic waves at very different moments, which suggested that a time-series visualization would be necessary to compare timing, magnitude, and duration across countries.

Early comparisons also showed that raw case and death totals were not ideal for world comparison because large-population countries tended to dominate the rankings even when their relative burden was lower. This insight led me to use per-million measures for the choropleth map so that countries could be compared more fairly on a population-adjusted basis.

Exploration of vaccination variables indicated that vaccination coverage needed to be shown alongside, rather than separately from, cases and deaths, because one of the project's goals was to help users visually explore whether later waves differed after vaccine rollout expanded. This motivated the decision to include a metric switch that lets users move between cases, deaths, and vaccinations while keeping the same interaction structure.

The EDA stage also helped determine the final dashboard components. Geographic patterns were best represented through a world map, differences in outbreak timing were best shown through a multi-country line chart, and exact country-level values were easiest to present in a small detail panel. These observations shaped the final design by linking each analytical need to a specific chart type.

Design Evolution:

At the beginning of the project, the dashboard design was much less clear than it looks now. The very first ideas split the views across different screens: one page focused only on the world map, and a separate page showed time-series charts of cases and deaths. The thought was that this would keep each page simple, but in practice it meant users had to keep switching back and forth to connect “where” and “when.” There were also early sketches with bar charts comparing countries at a single date and with lots of small line charts arranged in a grid for each continent. Those experiments made it obvious that too many separate charts or pages quickly became confusing and made it harder to see the big picture.

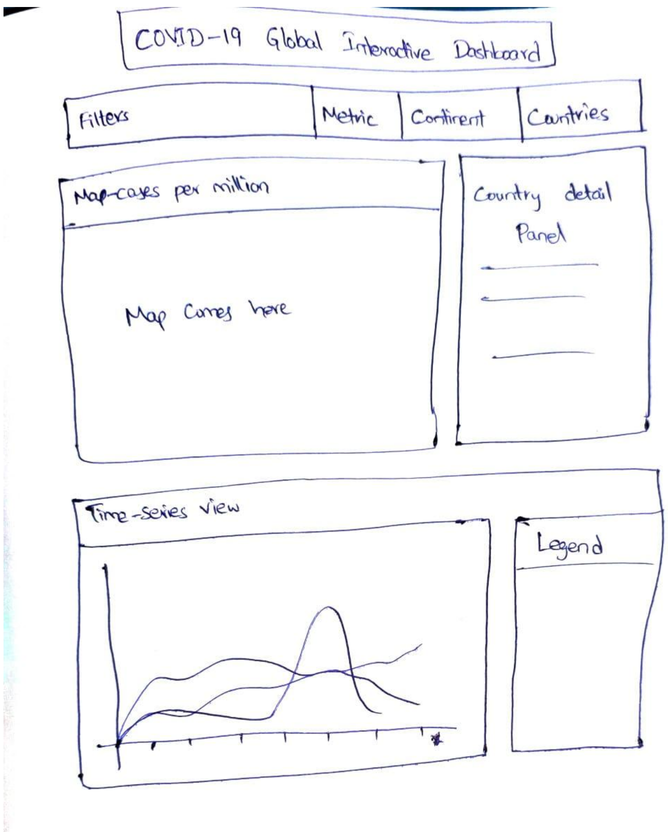

Figure 1: Early hand-drawn sketch of the dashboard layout showing filters, map, time-series view, legend, and country detail panel.

As the project progressed, the design shifted toward putting the most useful pieces together in one place. The map and the multi-country time-series chart moved onto the same screen, with a single country detail panel on the side to show exact values. This layout lets someone start with a global overview, then immediately see how a few selected countries evolved over time and finally check detailed numbers for one country without ever leaving the page. The interactions also became simpler over time: instead of lots of advanced options, the final version keeps only a small set of controls - metric, continent, and country selection—so the dashboard feels approachable for non-experts while still supporting the main questions about waves, relative burden, and vaccination patterns.

Visualization Design:

The final dashboard combines three coordinated components: a choropleth world map, a multi-country time-series chart, and a country detail panel. This design was chosen because each component answers a different part of the overall problem: the map provides a global spatial overview, the line chart shows temporal changes and wave patterns, and the detail panel gives exact summary values for the currently selected country.

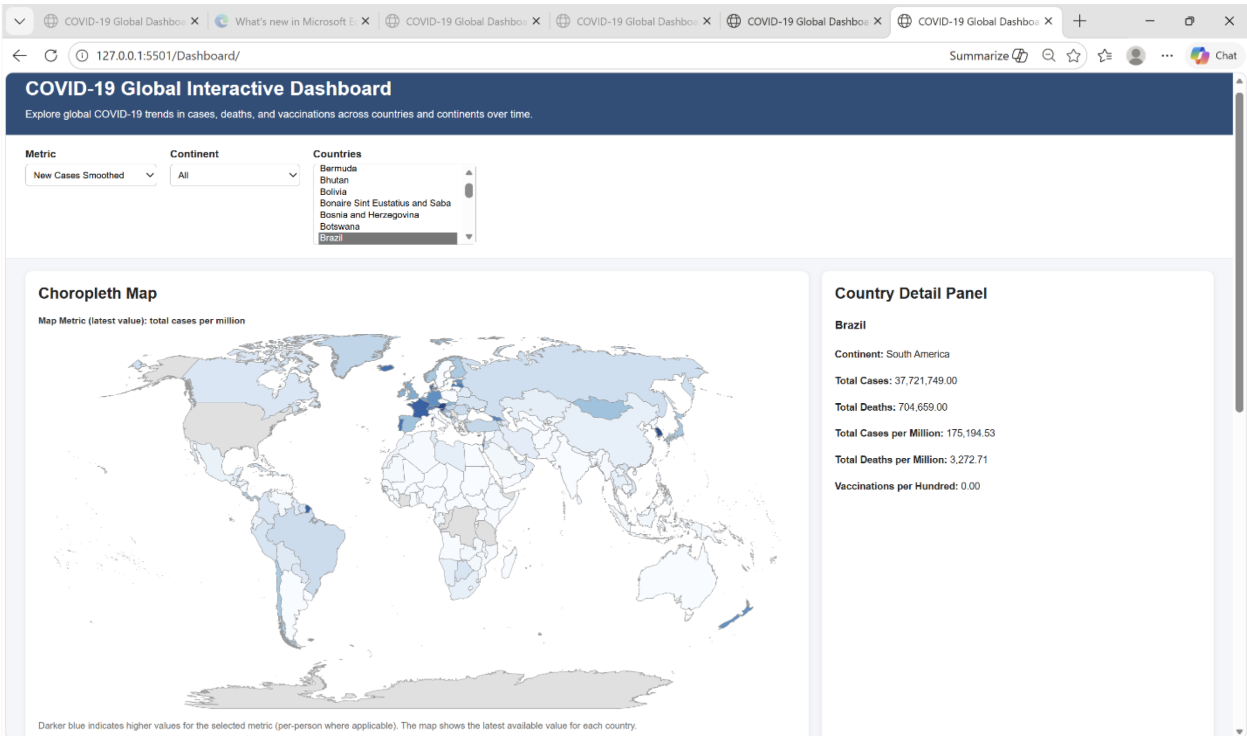

Figure 2: Overall COVID-19 Global Interactive Dashboard showing filters, choropleth map, and country detail panel.

The choropleth map was designed to support quick comparison of countries at the latest available point in time using normalized indicators such as total cases per million and total deaths per million. This follows common guidance for COVID-19 visualizations, which recommends showing trends with context and using population-aware measures rather than relying only on raw counts. A short explanatory note was also added below the map to clarify the meaning of color intensity and reduce ambiguity.

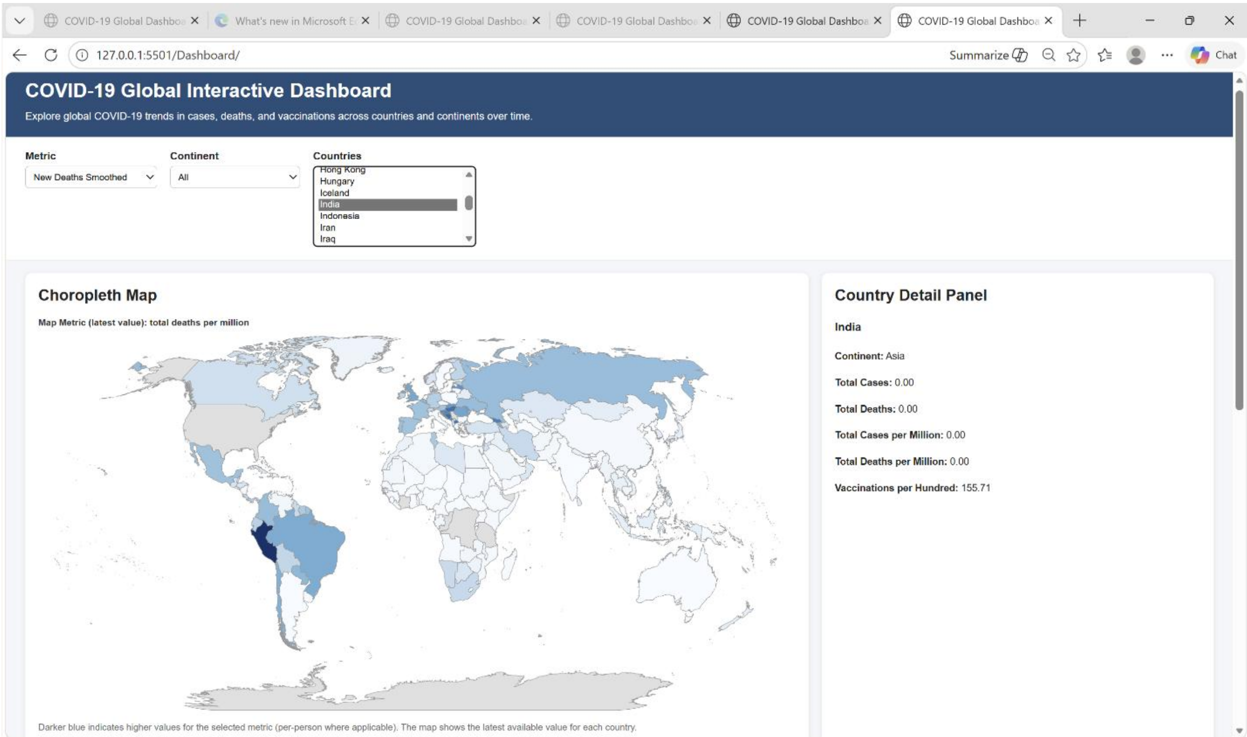

Figure 3: Choropleth map of total deaths per million with a selected country, illustrating geographic variation in COVID-19 mortality.

The multi-country time-series chart was included because one-day snapshots cannot explain how the pandemic evolved over time. By allowing users to compare selected countries on the same chart, the dashboard makes it possible to identify differences in wave timing, peak size, and post-vaccination trajectories. A legend and metric-specific y-axis labeling were used to make interpretation easier.

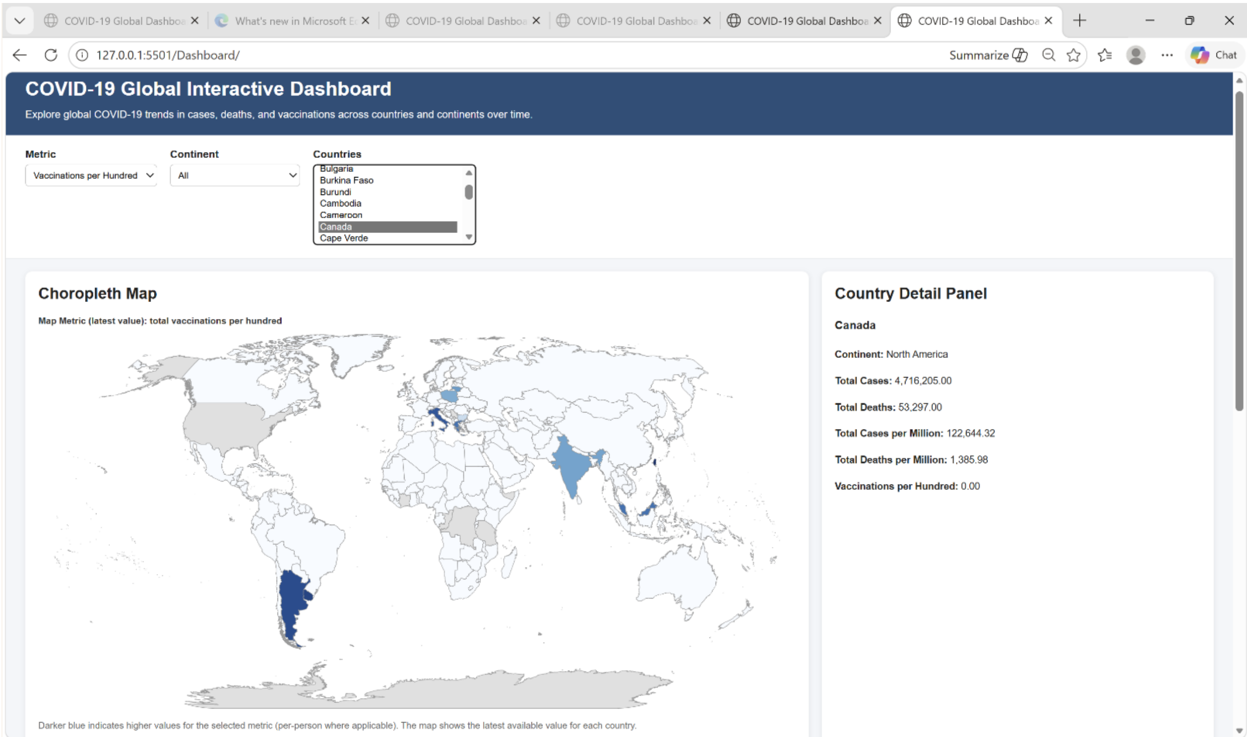

Figure 4: Multi-Country Time Series of new cases smoothed, enabling comparison of pandemic waves across countries.

Interactive filters were added for metric, continent, and country because dashboard best practices emphasize usability, comparison, and clarity. The metric filter supports switching between cases, deaths, and vaccination measures; the continent filter narrows the country list; and clicking a country on the map updates the detail panel and chart selection. Together, these controls help users move from broad global comparison to focused country-level exploration without leaving the same interface.

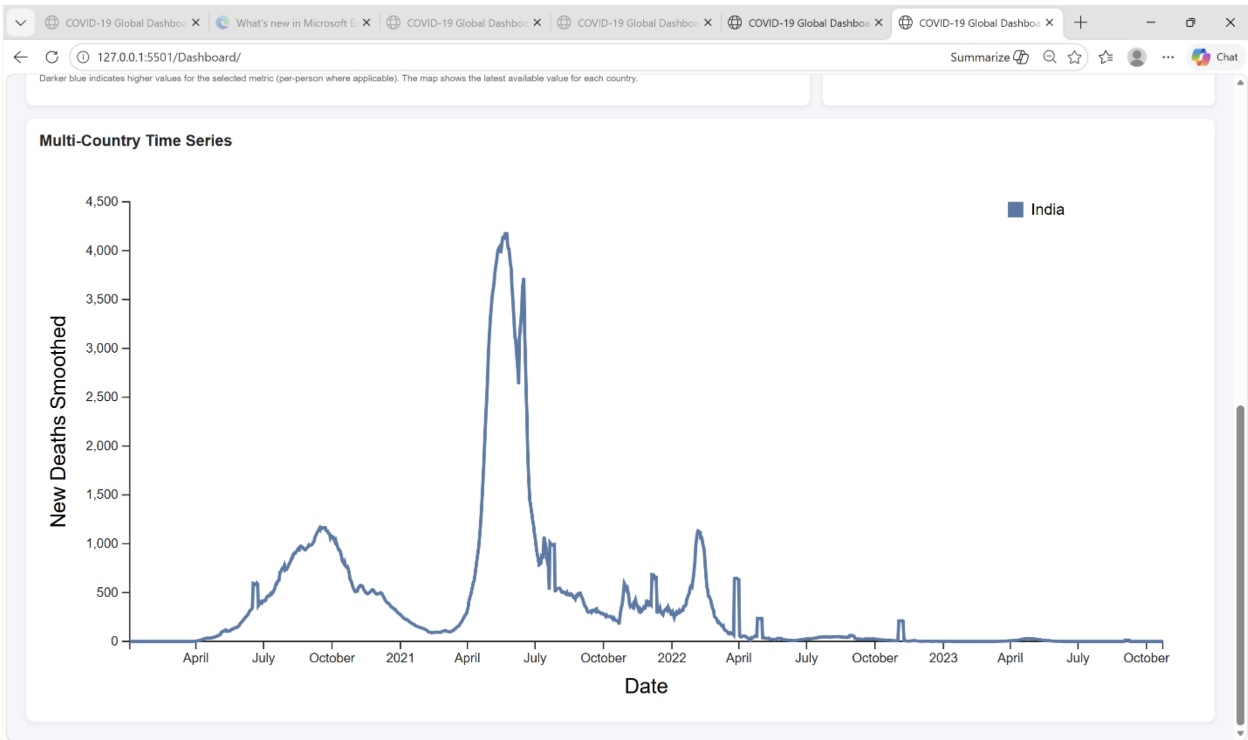

Figure 5: Vaccinations per hundred over time for one country, illustrating vaccine rollout and saturation.

Implementation:

The dashboard was implemented as a web-based interactive visualization using HTML, CSS, JavaScript, and the D3.js library. D3 was selected because it provides fine-grained control over data binding, scales, map projections, SVG rendering, and interactive event handling, which made it suitable for combining geographic and temporal views in a single interface.

The implementation process began with preprocessing the original Our World in Data COVID-19 dataset into smaller files optimized for the dashboard. Instead of loading the full raw dataset directly into the browser, I created a time-series file for charting (dashboard_data.csv), a latest-value snapshot for the choropleth map and country detail panel (latest_snapshot.csv), a continent summary file to support filtering (continent_summary.csv), and a country. geojson file for country boundaries. This structure reduced complexity in the frontend and improved interaction speed because only the necessary data were loaded for each component.

The interface contains three coordinated views. The choropleth map displays the latest available value of the currently selected metric for each country using a color scale, with darker blue indicating higher values. The multi-country line chart shows how the selected metric changes over time for chosen countries, making it possible to compare wave timing and intensity. The country detail panel presents exact summary values for the selected country, such as total cases, total deaths, and vaccination coverage.

Interactivity was implemented through several linked controls. The metric dropdown updates both the line chart and the map so users can switch between cases, deaths, and vaccinations without changing the rest of the interface. The continent filter narrows the country list, reducing clutter and supporting more focused comparison. The country selector allows manual selection of multiple countries, while clicking a country on the map updates the active selection and the detail panel. This linked-view behavior was important because it lets users move naturally between overview and detail, which is a common goal in dashboard design.

During implementation, I also made several interface refinements to improve clarity and usability. For example, the map metric label was moved outside the map graphic so it would not overlap with the visualization, and a short explanatory note was added under the map to clarify how color should be interpreted. These adjustments improved readability and made the final dashboard more polished and easier to understand.

Evaluation:

To evaluate the usefulness and usability of the dashboard, I planned a small task-based user test with 2 to 3 classmates or friends. A task-based approach is appropriate for dashboards because it measures whether users can answer questions by interacting with the visualization rather than only giving general opinions. This type of evaluation helps identify whether the layout, labels, and interactions support the intended analytical tasks.

Participants were asked to complete a small set of realistic tasks using the dashboard. Example tasks included: comparing COVID-19 waves between India and the United States to identify which country had the higher peak and when; using the map to find a European country with high deaths per million; and selecting a country to describe how vaccination per hundred changes in relation to cases or deaths over time. These tasks were chosen because they reflect the core analytical goals of the project: temporal comparison, geographic comparison, and linking vaccination trends to pandemic outcomes.

After completing the tasks, participants were asked a few short questions: what was easy to understand, what was confusing, and what suggestions they had for improvement. The evaluation focused on factors commonly used in dashboard usability studies, including ease of use, learnability, suitability for the task, and clarity of the interface. This made the evaluation simple enough to conduct informally while still being meaningful for design improvement.

The feedback was used to guide small refinements in the final design. For example, one improvement was separating the map metric label from the map itself, so the title no longer visually merged with the visualization. Another improvement was adding explanatory text below the map and clarifying labels across the interface. These changes show that the dashboard was not only implemented but also iteratively refined based on user experience, which strengthens the final design outcome.

In future work, the evaluation could be expanded with a larger sample size, timed tasks, or a short questionnaire to quantify user performance and satisfaction more systematically. However, even a small usability test is valuable because it reveals whether the dashboard communicates its message clearly to new users.

Discussion:

The dashboard provides an integrated view of global COVID-19 patterns by combining a choropleth map, a multi-country time series, and a country detail panel. Together, these components make it easier to see how the timing and intensity of waves varied across countries and continents, and to visually explore how vaccination coverage relates to later changes in cases and deaths. Users can, for example, identify countries with high deaths per million on the map and then use the time series to examine whether those countries experienced distinct peaks or prolonged waves.

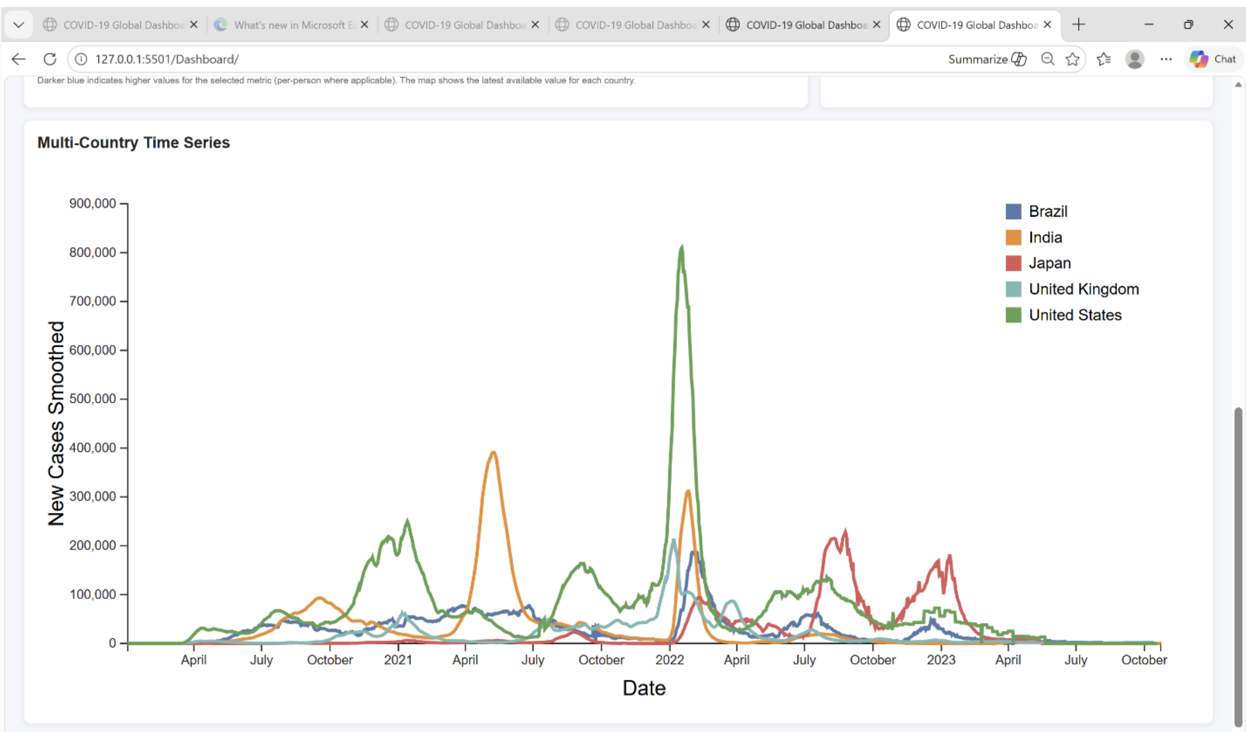

Figure 6: Multi-Country Time Series focused on a single country, highlighting distinct COVID-19 case waves over time.

The interactive filters also support nuanced analysis. Allowing users to switch between cases, deaths, and vaccination metrics means the same interface can answer different questions without overwhelming the viewer with too many charts at once. Continent-based filtering simplifies country selection, helping users focus on a region of interest while still enabling global comparison when needed. The consistent use of per-million and per-hundred metrics reduces the risk of misinterpreting raw counts, particularly for highly populated countries.

From a design perspective, the project highlights how coordinated views and careful encoding choices can help non-expert users navigate a complex dataset. Small interface decisions - such as moving titles outside the plots, adding short explanations, and aligning color scales across views - contribute to a more coherent and interpretable experience. Overall, the dashboard shows that a relatively small number of well-chosen visual components can support rich exploration when they are tightly integrated.

Limitations:

Despite its strengths, the dashboard has several important limitations. First, it depends entirely on the quality and completeness of the underlying COVID-19 dataset, which varies across countries due to differences in testing capacity, reporting practices, and definitions of cases or deaths. Apparent differences between countries may therefore reflect data artefacts as much as true differences in disease burden, which can mislead users if they are not aware of these issues.

Second, the dashboard is descriptive rather than causal. It allows users to observe that high vaccination coverage is associated with certain patterns in cases or deaths, but it does not control for confounding factors such as age structure, healthcare capacity, non-pharmaceutical interventions, or variant spread. As a result, the interface supports hypothesis generation but cannot be used to draw strong causal conclusions about the effect of any single policy or intervention.

There are also limitations in the scope of the metrics and interactions. The map currently shows only the latest available values for each country, which simplifies interpretation but hides the geographic evolution of the pandemic over time. Users must rely on the time series chart to study temporal change. In addition, the dashboard focuses on a small set of indicators and does not include hospitalizations, test positivity, ICU occupancy, or variant information, which could provide a more complete picture. Finally, the evaluation was based on a small number of informal tests rather than a large, systematic study, so the usability findings should be seen as preliminary and may not generalize to all user groups.

Conclusion:

This project set out to address a common problem with COVID-19 data: although large amounts of information are publicly available, most existing dashboards and charts make it hard for non-experts to compare countries, understand the timing of waves, or see how vaccination patterns relate to cases and deaths. The work focuses on this gap and presents an interactive dashboard that emphasizes clarity and comparability rather than simply displaying many plots. In doing so, the project turns a complex, messy dataset into a tool that supports meaningful exploration and insight for a general audience.

To achieve this, the project collects and cleans a global COVID-19 dataset, transforming raw counts into comparable per-capita measures such as cases per million, deaths per million, and people vaccinated per hundred. This preprocessing step is crucial because it allows users to compare countries of very different population sizes in a fair way instead of being misled by raw totals. The data is also organized so that it can be filtered easily by country, continent, metric, and time, which directly supports the interactive features of the dashboard.

On the visualization side, the dashboard uses three coordinated views: a choropleth map to show the global situation briefly, a multi-country time series to reveal how pandemic waves evolved over time, and a country detail panel to show focused statistics for a selected country. These views are linked through interactions, so choices made in one component (for example, selecting a continent or a country) update the others. This coordinated design solves the problem of having to switch between many disconnected charts and instead provides a coherent story about how COVID-19 unfolded across regions.

The interface also introduces simple but effective interaction mechanisms—such as dropdowns or buttons to select cases, deaths, or vaccination metrics; filters by continent; and selection of specific countries—to let users tailor the dashboard to their questions without being overwhelmed. By limiting the number of metrics shown at once and keeping visual encodings consistent (for example, using the same color scales and time axis), the design reduces cognitive load and makes the interface more approachable for non-expert users. These choices directly address the initial problem of dashboards being confusing and hard to interpret.

In the end, the project demonstrates that careful data preparation, thoughtful choice of visual encodings, and user-centered interaction design can turn complex pandemic data into an accessible tool. The dashboard is not intended to answer causal questions or replace expert epidemiological analysis, but it does solve a specific, important problem: helping everyday users clearly compare COVID-19 patterns across countries and continents and visually explore how vaccination levels relate to cases and deaths over time. This contribution shows how interactive visualization can bridge the gap between raw public health data and understandable, interpretable insight.

Future Work:

There are several promising directions to extend and improve this COVID-19 dashboard. A first step would be to enrich the temporal aspect of the map by adding a time slider or animation controls, allowing users to watch how cases, deaths, and vaccination coverage spread and changed across the world over time. This would directly address the current limitation that the map shows only a single snapshot and would help users connect geographic patterns with the timing of pandemic waves.

Another important extension is to incorporate additional metrics that give a more complete picture of the pandemic. Examples include hospitalizations, ICU occupancy, test positivity rates, and information about dominant variants. Adding these indicators would allow users to explore questions such as how healthcare systems were stressed during specific waves or how new variants affected case and death trajectories. Any additions would need careful interface design to avoid cluttering and maintain the dashboard's focus on clarity and interpretability.

Future work could also focus on improving the interactivity and analytical capabilities of the dashboard. This might include enabling users to define and save custom country groups (for example, "high-income countries" or "neighboring countries"), adding simple statistical summaries (such as moving averages or percentage changes over specified periods), or allowing users to annotate charts with key events like lockdowns or vaccine rollouts. These features would help users move from simply viewing data to actively constructing and communicating their own insights.

A more rigorous evaluation of the dashboard is another key direction. Conducting a structured usability study with a larger and more diverse set of participants would provide stronger evidence about which design choices are effective and where users still struggle. Task-based evaluations and think-aloud sessions could reveal specific interaction patterns, misunderstandings, or missing features, directly informing the next iteration of the design.

Finally, the underlying design principles of this project can be generalized beyond COVID-19. Future work could explore applying the same approach - coordinated views, per-capita metrics, and simple, meaningful interactions - to other public health and societal datasets, such as influenza surveillance, air quality and pollution, or chronic disease indicators. Doing so would demonstrate how the techniques developed in this project can support transparent, accessible data communication in a wide range of real-world contexts.

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